

Yu-Wei Fan

yf9172@princeton.edu | (+1) 609-436-8519

Research Interests

(AI-assisted/Agentic) Formal verification, automated reasoning, and electronic design automation (EDA).

Education

Ph.D. in Electrical and Computer Engineering	2024–2029 (expected)
<i>Princeton University, USA</i>	Advisor: Prof. Sharad Malik
M.A. in Electrical and Computer Engineering	2026
<i>Princeton University, USA</i>	Advisor: Prof. Sharad Malik
M.S. in Electronics Engineering	2024
<i>National Taiwan University (NTU), Taiwan</i>	Advisor: Prof. Jie-Hong Roland Jiang
B.S. in Electrical Engineering	2022
<i>National Taiwan University (NTU), Taiwan</i>	

Research Experience

Research Assistant	2024–Present
<i>Princeton University, Sharad Malik's Group</i>	Advisor: Prof. Sharad Malik
• Researching agentic formal hardware verification for both model checking and Lean theorem proving. • Researching automated optimization of taint propagation logic for hardware IFT. • Researching hardware model checking algorithms for information-flow properties.	
Research Assistant	2020–2024
<i>National Taiwan University, Applied Logic and Computation Lab</i>	Advisor: Prof. Jie-Hong Roland Jiang
• Researching stochastic Boolean satisfiability solving and its extensions. • Researching reverse engineering of gate-level arithmetic circuits.	
Adjunct Research Assistant	2022–2023
<i>Institute of Information Science, Academia Sinica</i>	Advisor: Dr. Yu-Fang Chen
• Applied software testing techniques, including concolic testing, to DNN models.	

Publications

* indicates equal contribution

Preprints and Manuscripts Under Review

- [P1] **Yu-Wei Fan**, SooHyuk Cho, Aarti Gupta, and Sharad Malik. “Agentic-IC3: Enabling Semantic Proof Search in IC3 Model Checking”. Under review. 2026. arXiv: 2609.27162.
- [P2] Christine Guo, **Yu-Wei Fan**, SooHyuk Cho, Aarti Gupta, and Sharad Malik. “Resolving Signal Correlations for Precise Hardware Information-Flow Tracking”. Under review.
- [P3] **Yu-Wei Fan**^{*}, Yuheng Yang^{*}, Christine Guo, SooHyuk Cho, Thomas Bourgeat, Mengjia Yan, and Sharad Malik. “Efficient Hardware Information-Flow Tracking for Pre-Silicon Security Testing”. Under review. 2026. arXiv: 2609.06791v1.
- [P4] Long-Hin Fung, Che Cheng, **Yu-Wei Fan**, Tony Tan, and Jie-Hong Roland Jiang. “2-DQBF Solving and Certification via Property-Directed Reachability Analysis”. Under review.

Conference Papers

- [C1] Tingzhen Dong^{*}, Kunpeng Wang^{*}, Yuheng Yang, **Yu-Wei Fan**, Qinhan Tan, Thomas Bourgeat, Sharad Malik, and Mengjia Yan. “Interplay of Efficient Model Checking and Secure Processor Design: A Case Study on Secure Speculation”. In: *IEEE Symposium on Security and Privacy (SP)*. 2026.
- [C2] Kuo-Wei Ho, Jie-Hong Roland Jiang, Alan Mishchenko, Sean A. Weaver, and **Yu-Wei Fan**. “Hierarchical Boundary Recovery: Overcoming Synthesis Obscurity via SAT Sweeping”. In: *The Chips To Systems Conference (DAC)*. 2026.
- [C3] Qinhan Tan, Akash Gaonkar, **Yu-Wei Fan**, Sharad Malik, and Aarti Gupta. “SecIC3: Customizing IC3 for Hardware Security Verification”. In: *Design, Automation and Test in Europe Conference (DATE)*. **Best Paper Nomination**. 2026.
- [C4] **Yu-Wei Fan**, and Jie-Hong Roland Jiang. “Unifying Decision and Function Queries in Stochastic Boolean Satisfiability”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2024.

[C5] Long-Hin Fung, Che Cheng, **Yu-Wei Fan**, Tony Tan, and Jie-Hong Roland Jiang. “2-DQBF Solving and Certification via Property-Directed Reachability Analysis”. In: *Formal Methods in Computer-Aided Design (FMCAD)*. **Best Paper Award**. 2024.

[C6] **Yu-Wei Fan**, and Jie-Hong Roland Jiang. “SharpSSAT: A Witness-Generating Stochastic Boolean Satisfiability Solver”. In: *AAAI Conference on Artificial Intelligence (AAAI)*. 2023.

[C7] Kuo-Wei Ho, Shao-Ting Chung, Tian-Fu Chen, **Yu-Wei Fan**, Che Cheng, Cheng-Han Liu, and Jie-Hong Roland Jiang. “WolfEx: Word-Level Function Extraction and Simplification from Gate-Level Arithmetic Circuits”. In: *International Conference on Computer-Aided Design (ICCAD)*. 2023.

Awards and Honors

2026	Princeton ECE Pramod Subramanyan *17 Early Career Graduate Award
2024	Institute of Information and Computing Machinery Master Thesis Award, Honorable Mention
2024	IEEE Taipei Section Master Thesis Award
2024	Taiwanese Association of Artificial Intelligence Master Thesis Award, Honorable Mention
2024	NTU Graduate Institute of Electronics Engineering Master Thesis Award

Teaching Experience

Teaching Assistant, Contemporary Logic Design, Princeton University	Fall 2025, 2026
Teaching Assistant, Introduction to Electronic Design Automation, NTU	Spring 2023, 2024
Teaching Assistant, Switching Circuits and Logic Design, NTU	Fall 2022

Skills

Programming Languages	C/C++, Python, Verilog/SystemVerilog, Rust
Verification	SAT/SMT solvers (Z3/CVC5, CaDiCaL), model checkers (rIC3, AVR/Pono), Berkeley ABC Logic Synthesis and Verification System
EDA	Yosys Open Synthesis Tools
Languages	Mandarin (Native), English (Proficient)